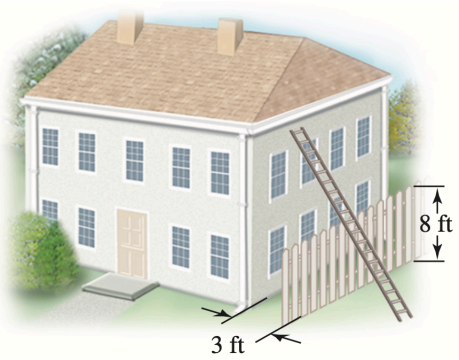
Name:_____

Here is an assortment of optimization problems involving geometry.

1. An 8-foot-tall fence runs parallel to the side of a house 3 feet away. What is the length of the shortest ladder that clears then fence and reaches the house? (Assume that the vertical wall of the house and the horizontal ground have infinite extent.)



2. A right circular cylinder is inscribed in a sphere of radius R. Find the largest possible volume of such a cylinder. Include a properly-labeled diagram, show all work, and justify your answer. (Relevant volume formulas: $V_{\text{sphere}} = \frac{4}{3}\pi r^3$, $V_{\text{cyl}} = \pi r^2 h$.)

3. A 2 ft piece of wire is cut into two pieces. One piece is bent into a square and the other is bent into an equilateral triangle.

   (a) Where, if anywhere, should the wire be cut so that the total area enclosed by both is minimized?

   (b) Where, if anywhere, should the wire be cut so that the total area enclosed by both is maximized?

4. A trough for holding water is formed by taking a piece of sheet metal 60 cm wide and folding the 20 cm on either end up as shown below. Determine the angle θ that will maximize the amount of water that the trough can hold.

